

ROHAN MISTRY

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SKILLS

Languages: Python, Java, JavaScript, TypeScript, Go, C/C++, HTML/CSS, SQL, R, Svelte

Frameworks & Libraries: React, Node.js, Next.js, Nest.js, Kotlin, FastAPI, GraphQL, TensorFlow, PyTorch, Pandas, NumPy

Tools & Platforms: Git, Docker, AWS, PostgreSQL, MySQL, Kubernetes, Maven, Tomcat, Tableau, Figma

EXPERIENCE

Full-Stack Software Engineer Intern | GoFundMe

May 2026 — Present

- Co-led end-to-end development of leaderboard share assets feature for GoFundMe Peer-to-Peer fundraisers across production codebases, from data layer design and release gating to operational observability metrics product analytics.
- Shipped features and production fixes across **React/TypeScript** frontends, **NestJS/GraphQL** API layer, and **Kotlin** backend.
- Supported technical research for whitelabeling initiative and conducted cross-team QA testing for other product features.

Research Data Engineer | Humanity Unleashed Initiative

Sep 2024 — Jan 2025

- Engineered automated **Python** data pipelines to extract and standardize 800,000+ economic time series from FRED API and government databases, creating unified JSONL datasets for **multimodal AI model training**.
- Collaborated with research teams on foundation model architectures and time series forecasting, conducting literature reviews and implementing scalable data infrastructure using cloud storage.

Software & Data Intern | MKS Instruments

Oct 2023 — Sep 2024

- Developed and deployed automated **Python** web scraping pipelines to streamline market research processes, reducing manual data collection by 75% and accelerating competitive insights.
- Delivered high-quality datasets, managed **SAP** inventory analyses, and created marketing collateral for new product launches.

PROJECTS

NFL Mock Draft Simulator

React, FastAPI, Python, HTML/CSS, JavaScript, PostgreSQL, SQLAlchemy, Netlify, Render, Supabase

- Built full-stack web application simulating the NFL Draft with real-time team management and drafting logic.
- Automated CPU drafting using weighted algorithms for positional needs and player rankings.
- Integrated **React** frontend, **FastAPI** backend, and **PostgreSQL** in cloud architecture supporting RESTful APIs, CRUD operations, and reliable data synchronization via connection pooling.

Fabflix

Java, JavaScript, MySQL, jQuery, AJAX, Docker, AWS, Maven, Tomcat, JDBC, Kubernetes, JMeter

- Engineered cloud-native platform to browse/purchase from catalog of 15,000 films.
- Developed robust architecture with **Tomcat**, **Java** servlets, **Maven**, and **MySQL**, implementing full-text search with autocomplete, session-based checkout, and secure authentication with HTTPS, AJAX, and reCAPTCHA.
- Optimized performance with MySQL connection pooling, Apache load balancing, and JMeter performance testing.
- Deployed **Kubernetes** cluster on **AWS (EC2)** using **Docker** containers for horizontal scalability.

Traffic Signal Control Using Reinforcement Learning

Python, PyTorch, RLlib, Pandas, NumPy, JavaScript, CityFlow

- Developed AI-driven multi-agent RL framework using MAPPO algorithms to optimize traffic signals, reducing congestion and increasing lane throughput in **CityFlow** simulations.
- Designed custom state/action representations and dynamic reward functions.
- Leveraged **Ray RLlib** for training/tuning, achieving measurable gains in traffic flow metrics on real-world datasets.

StrataScratch Exploratory Data Analysis

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, DeepNote, Alteryx Designer

- Performed end-to-end EDA pipeline on market data, cleaning and visualizing data with **Python** libraries to reveal actionable insights on supply/demand and pricing trends. Trained decision tree models with 90% predictive accuracy using **Scikit-learn**.
- Awarded *Best Analysis* at UC Irvine Datathon 2024. Collaborated via DeepNote/Alteryx for analytics workflows.

EDUCATION

University of California, Irvine

2021 — 2025

B.S. Computer Science — Specialization: Intelligent Systems

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Machine Learning & Data Mining, Database Management, Software Engineering, Operating Systems, Compilers, Information Retrieval

CERTIFICATIONS

AWS Certified Solutions Architect — Associate (SAA-C03) | Amazon Web Services

Mar 2026